Web Filtering Configuration on FortiGate Firewall

WRITTEN BY

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Lab Goal

In this lab, we simulated a school filter configuration, where we used a FortiGate firewall to block all traffic that can be deemed inappropriate for students to access, whilst also allowing educational websites through the filter. To do this, we needed to filter traffic through DNS filtering, Web filtering, and Application filtering.

Rationale behind filters.

As the purpose of this lab is to create a filter for schools, it is important to understand the rationale behind what we are banning and what we are allowing.

Banned sites

Illegal websites:

Students should not have access to

Drug abuse

Hacking

illegal information

discriminatory websites

websites that display explicit violence

extremist groups

proxy avoidance websites

plagiarism

child sexual abuse

terrorism

crypto mining

gambling

technology that can alter the student's computer

weapon sales

marijuana

tobacco

The school will be liable for all the damage that the student can cause from these websites. Thus, it is necessary to block these websites to ensure that students do not have access to resources that can damage the school's reputation or the student's future. Some of these sites are simply not appropriate for anyone (especially students) to access at all and need to be blocked.

Inappropriate content:

adult materials

nudity and risqué

pornography

dating

any suggestive material that may be related to pornography

Students should not be able to access these websites on school property such as laptops. Allowing them to access these would be irresponsible and ruin the school's reputation.

Security Risks

Malicious websites

Phishing

Spam URLs

Dynamic DNS

Newly Created Domains

Newly Registered Domains

Information and Computer Security

Remote Access

Freeware and Software Downloads

These websites have either been reported to be malware or phishing sites or visiting these sites pose a high risk due to how unknown they are.

Distractions

Brokerage and Trading

Games

Social Networking

Shopping

Web Chat

Instant Messaging

Newsgroups and Message Boards (like reddit)

Content Servers

Auction websites

Real Estate

URL Shortening wegpages

Resturaunt and Dining

Access to these sites would encourage students to get off task during class. Blocking messaging services other than the ones used by the school (e.x teams) can allow inappropriate content or plagiarism to be sent. If we only allow the school-provided messaging services, the school will have an easier time tracking these inappropriate or academically dishonest messages.

Privacy issues and Personal information

Advertising websites

Personal Privacy websites (like banking and health)

Finance and Banking

Students shouldn't be able to access sensitive information from a school computer. If a student were to have their information compromised from either a bystander or a security issue from the laptop, the school may be held liable for anything that happens. Advertisements should also be blocked because personalized ads may be a privacy issue for students logging into the internet on their personal accounts.

Allowed Sites

Educational sites

Sex Education (sexeducationforum.org.uk)

Arts and Culture (goodreads.com)

Education (nationalgeographic.com)

Health and Wellness (Health or medical services, self-help groups like webmd.com)

Job Searching (indeed.com)

Medicine(cvs.com)

News and Media (cnn.com)

Political Organizations (gop.com)

Reference sites (sites that offer general reference data libraries like wikipedia.com)

Global Religion (biblegateway.com)

Society and Lifestyles (sites that deal with general daily life issues)

Sports (espn.com)

Travel (booking.com)

Personal Vehicles (toyota.com)

Folklore (like UFOs, fortune telling)

Digital Postcards (123greetings.com)

Child Education (websites for educating children under 12)

Personal Websites and Blogs(blogspot.com)

Search Engines and Portals (google.com)

Government and Legal Organizations (whitehouse.gov)

Information Technology websites (microsoft.com)

Armed forces (military.com)

Web Based Applications (googledocs)

Charitable organizations (theliveyoucansave.org)

Web Analytics(openwebanalytics.com)

Online meetings(gotomeeting.com)

These sites provide educational benefits and it is unlikely for students to become distracted by these sites. Websites that appear here can be used for educational purposes by teachers and staff and can appear as resources or research projects.

Background Information

Fortigate has 3 types of filtering between webpages and domains: Application filtering, DNS Filtering, and Web Filtering.

Web filtering will block webpages based on a URL. This is particularly useful in school because it allows admins to block distracting URLs or potentially dangerous URLs from student's access. This is possibly the most important filter for students because it both generally blocks students from accessing dangerous websites, protecting them from the most common forms of malware, but also can make sure students aren't getting off task in class. Fortigate contains a database which can allow blocking URLs as a category such as Malware or Drugs/illegal substances, which adds to the simplicity in setting up a fortigate web filter. An example of this is any webpage on Facebook would be blocked because the domain facebook.com is blocked itself. DNS filtering will block webpages based on the IP address that is used to reach the webpage, which is layers under accessing the webpage. Thus, DNS filtering can filter based on known unwanted IP addresses, which can filter out spam or redirects from these IPs. Thus, these will completely block access to domains that have their IP address blocked. While this may be like Web Filter where an admin may block facebook.com/login but allow access to facebook.com/home, since both URLs come from the same IP address, then both URLs will be blocked. However, its only possible to use this filter when the fortigate firewall is able to access the fortiguard DNS servers, and thus can only be used if access to external DNS servers is allowed. This is useful in filtering out IP addresses that have been Application-level filtering is an incredibly broad level filtering of the highest level. Fortigate uses Application filtering to analyze web traffic and determine which specific service is being used such as FTP or HTTP, even if the traffic doesn't use the standard ports or protocols. This is particularly useful in cases such as preventing unwanted malware or risky applications that may present security issues with the ports they are connected to.

To apply these filters, the admin needs to bind the created filters to a created web policy, usually one that allows traffic to enter and leave the network and onto the network. A web policy to enter the network is needed along with a policy to leave the network because some websites simply forward their traffic to the network and take little user traffic to the website. This means that if the admin were to only create a policy exiting the network and not back in, any traffic sent back to the user will not be filtered because there is no firewall policy that is also blocking the traffic to come back in.

While websites may be blocked by fortigate, the webpage displayed can give confusion on whether the filter worked or not. The way Fortigate blocks websites Is by replacing the xwebsite's certification with its own. However, this can cause problems when the user's computer does not trust the certificate that is replacing these webpage's certificates

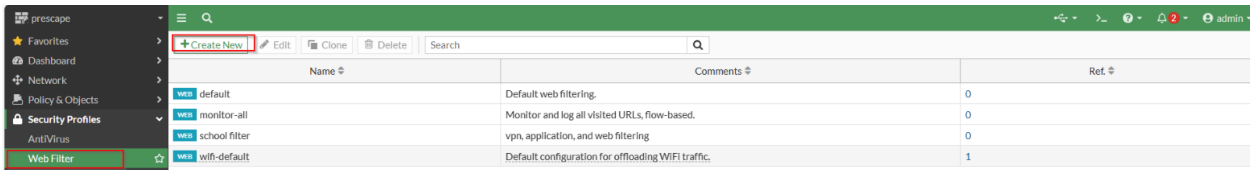
which can result in webpages showing a certificate error instead of a blocking error. To fix this, the user needs to download the Fortigate certificate off of the GUI which will then allow the computer to trust the firewall's certificates.

Since all filters will go through the fortigate's firewall, the GUI can display logs of any traffic that has been blocked or allowed. Thus, any troubleshooting in the Fortigate GUI through the web filters should be done there.

Lab Commands

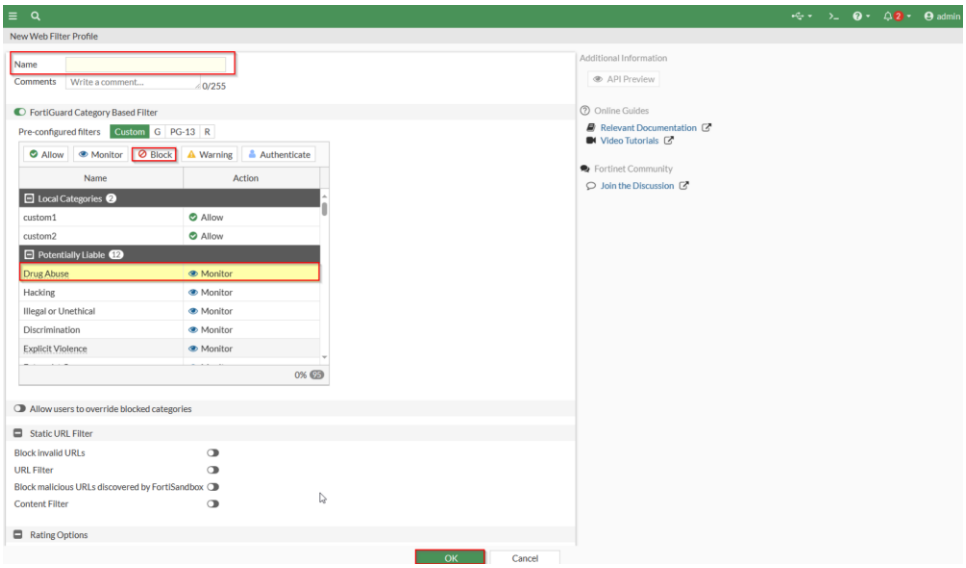
1.

Enter Web Filter and press "Create New"



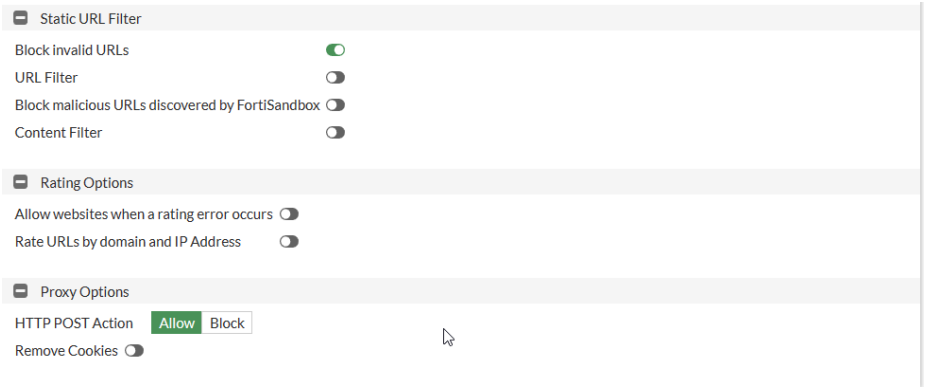
2.

Create a name and select each subject and in the top configuration menu, select which security method to choose. For example, select drug abuse and then select block.



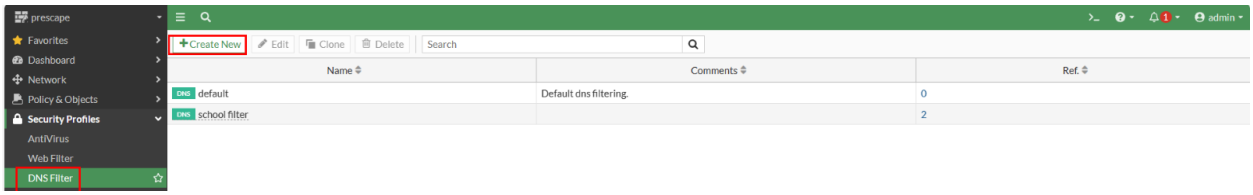
3.

underneath the Static Url Filter section, enable Block invalid URLs then press OK



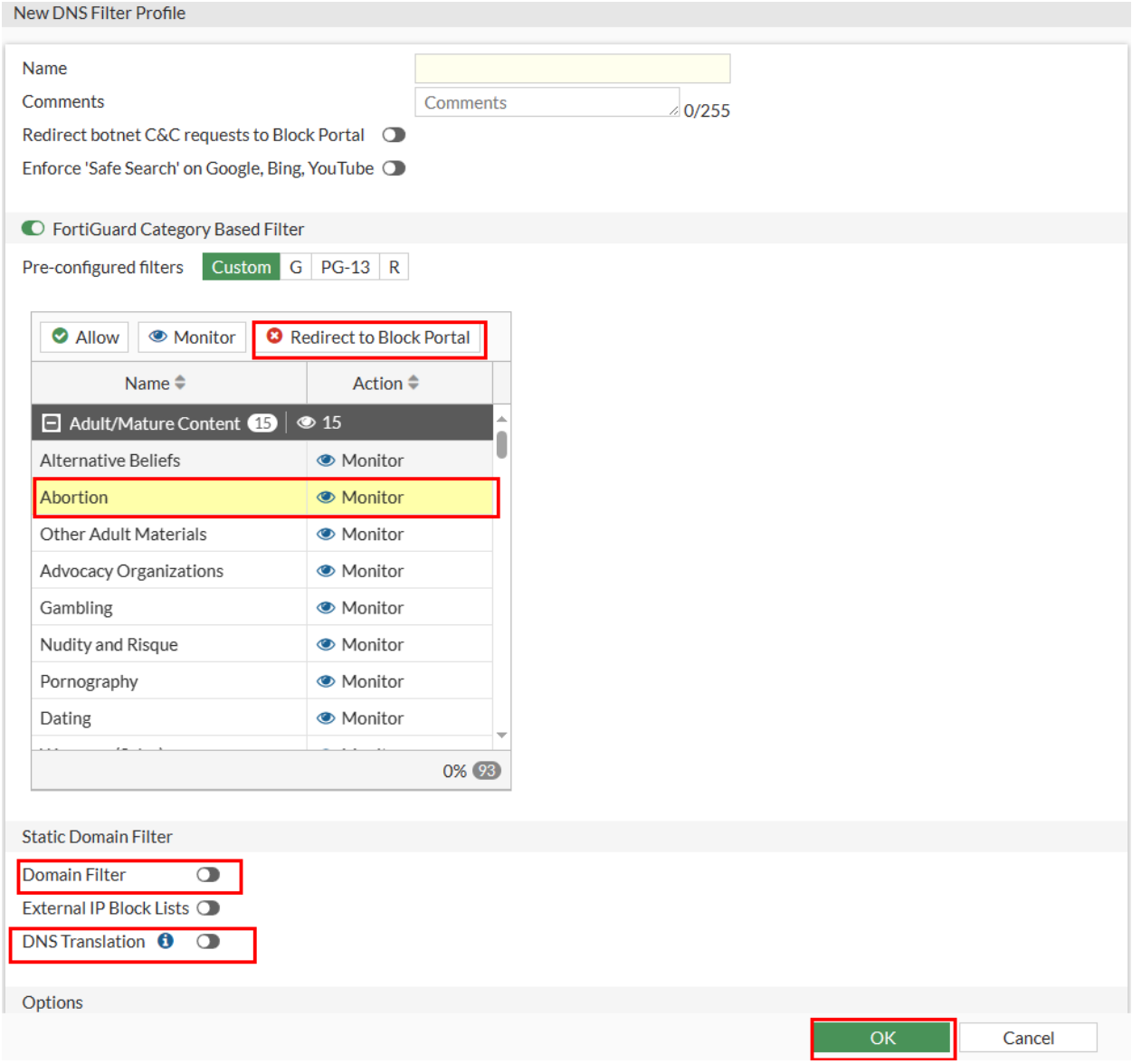
4.

enter security profiles > DNS filtering and select new



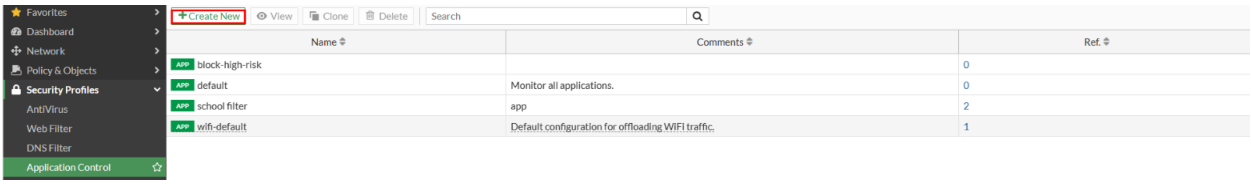
5.

Block any unwanted websites and turn on domain filter and dns translation. Then press ok.



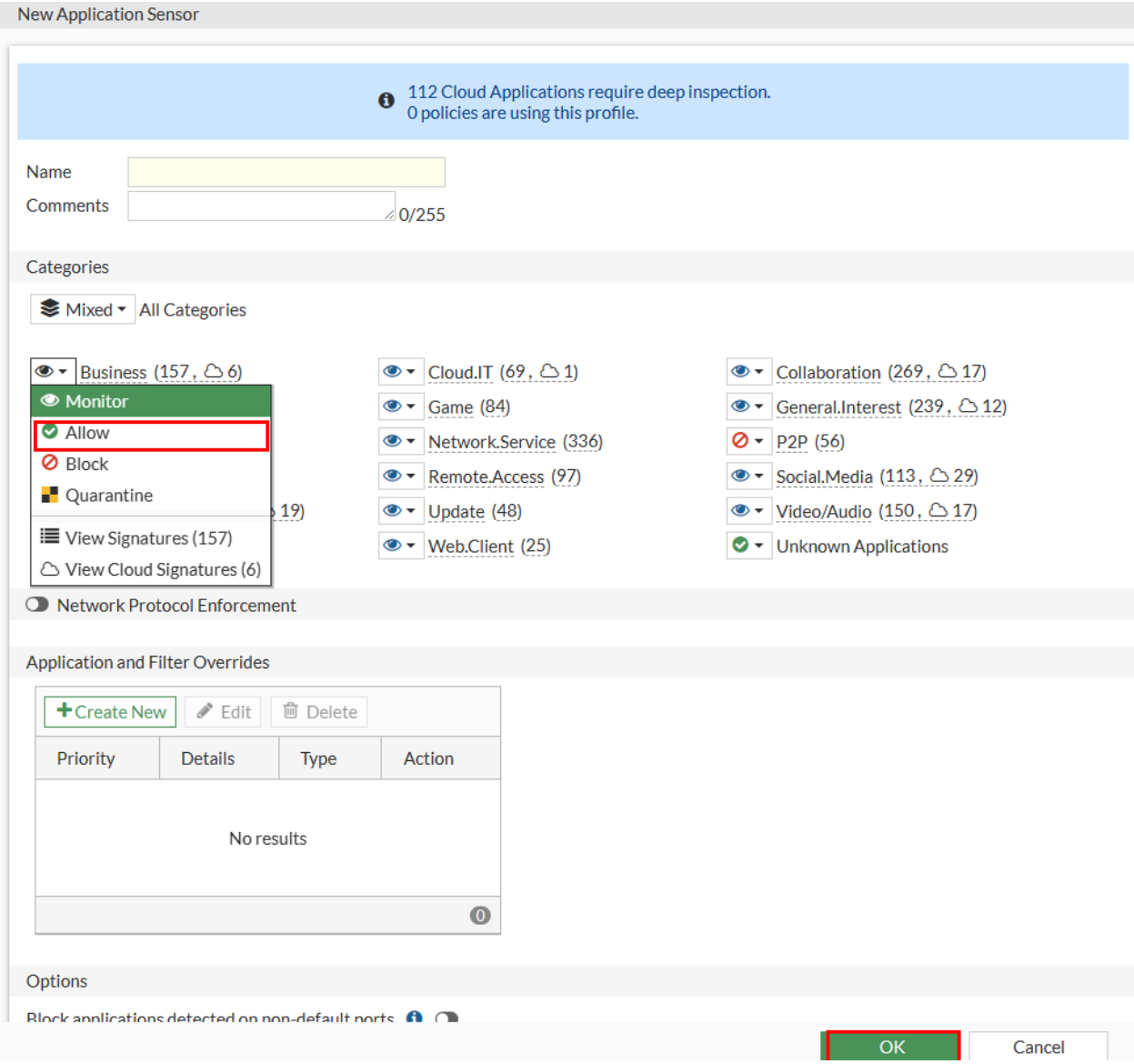
6.

Enter Security Profiles > Application Control and select “create new”



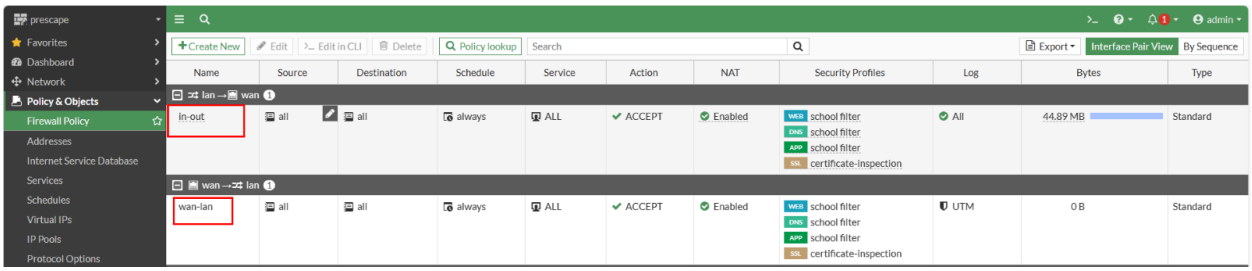
7.

Enter each category and block unwanted categories.



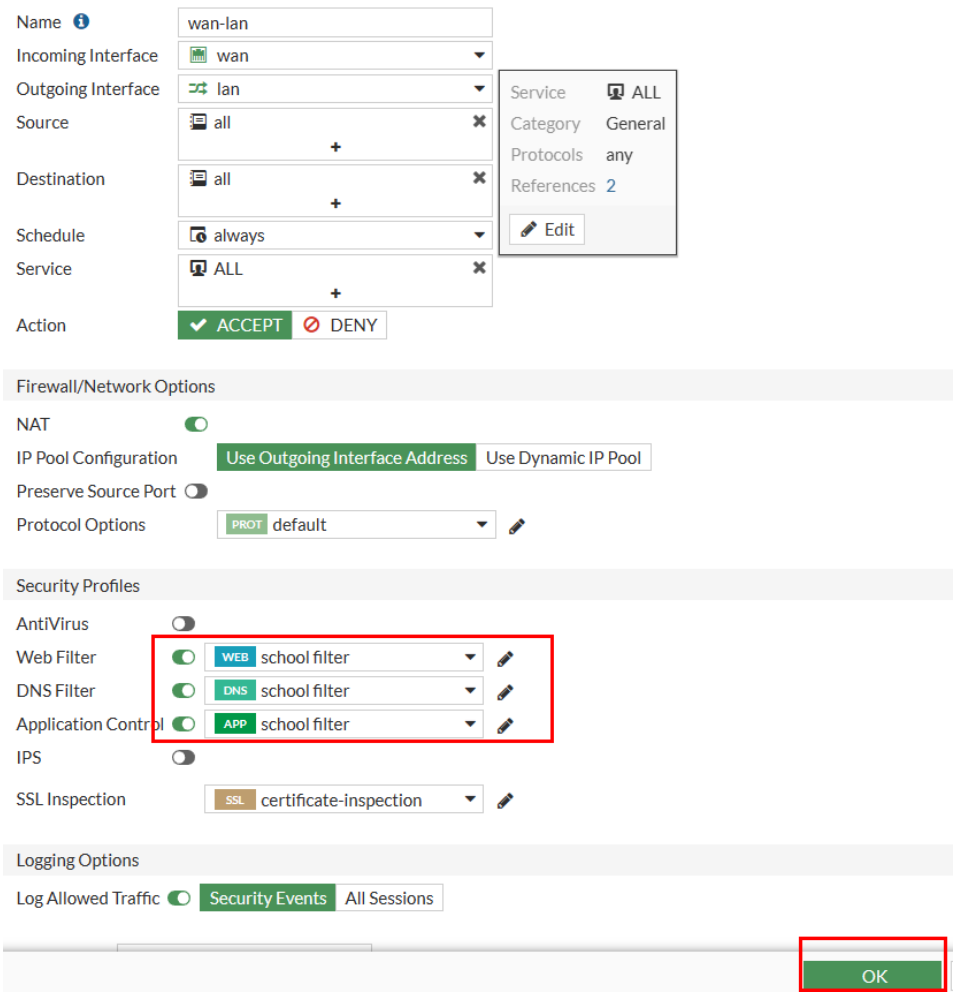
8.

Enter Policy and Objects > Firewall policy and create two policies. One to enter the lan and one to exit the lan.



9.

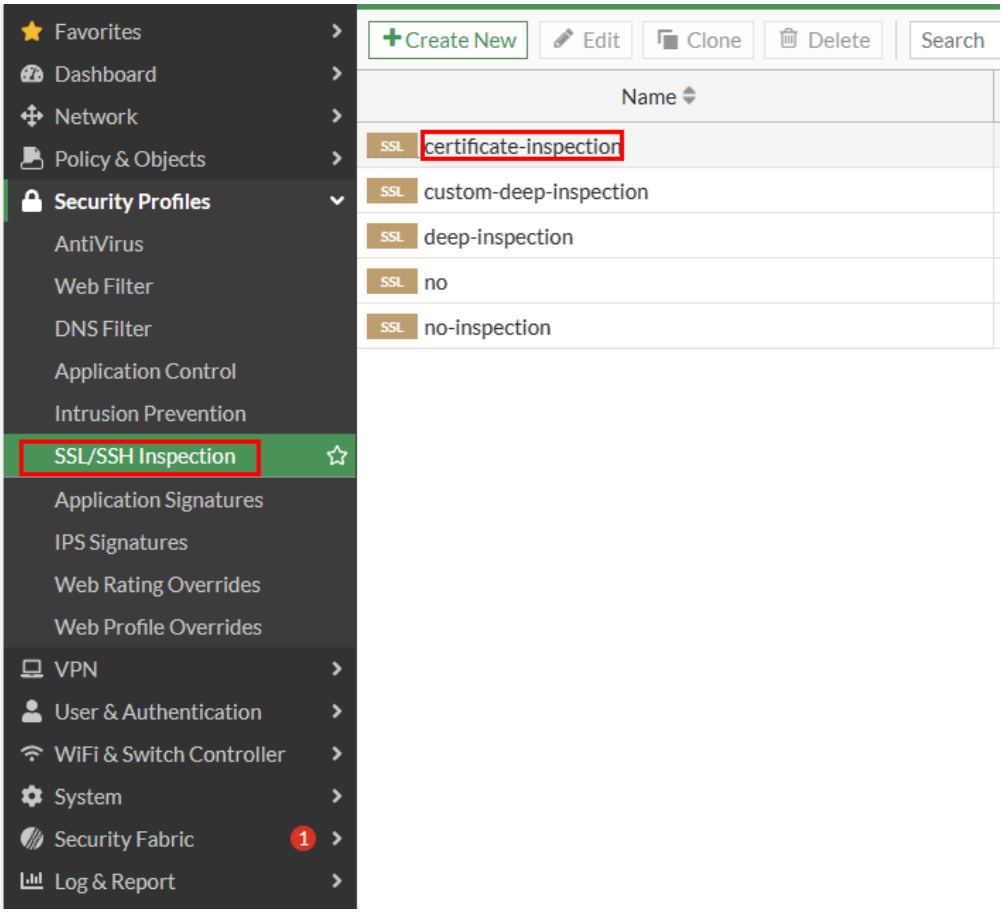
Enter both policies and enable web filter, dns filter, and application control. Then select OK.



Downloading SSL Certificate

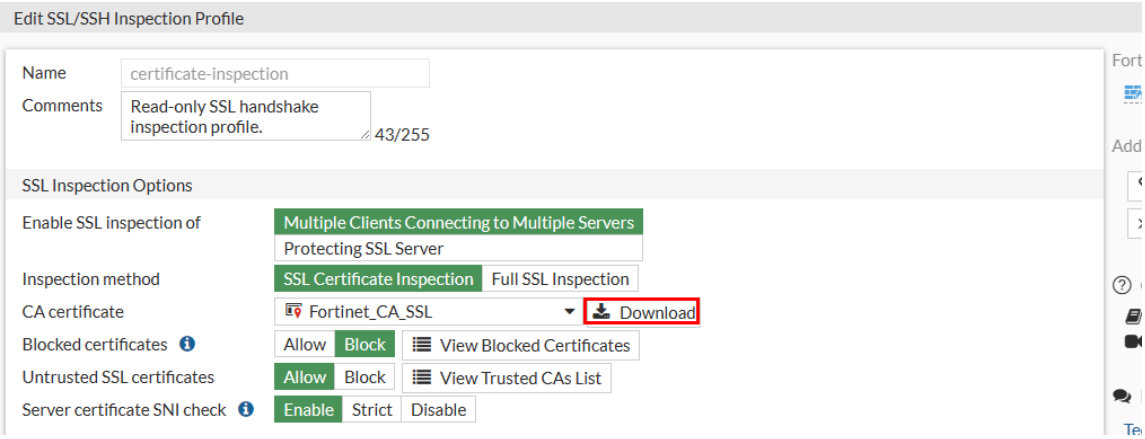
1.

Enter Security profiles > SSL/SSH Inspection and enter whichever certificate inspector you are using.



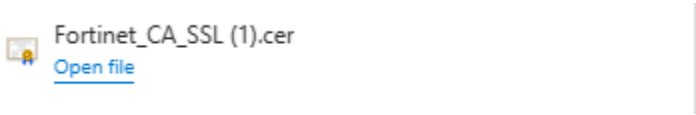
2.

Select Download



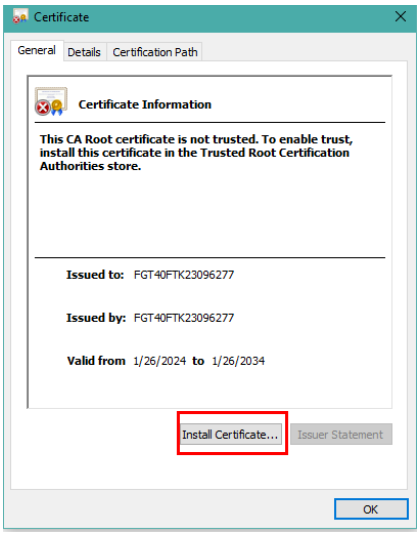
3.

Open the certificate file downloader



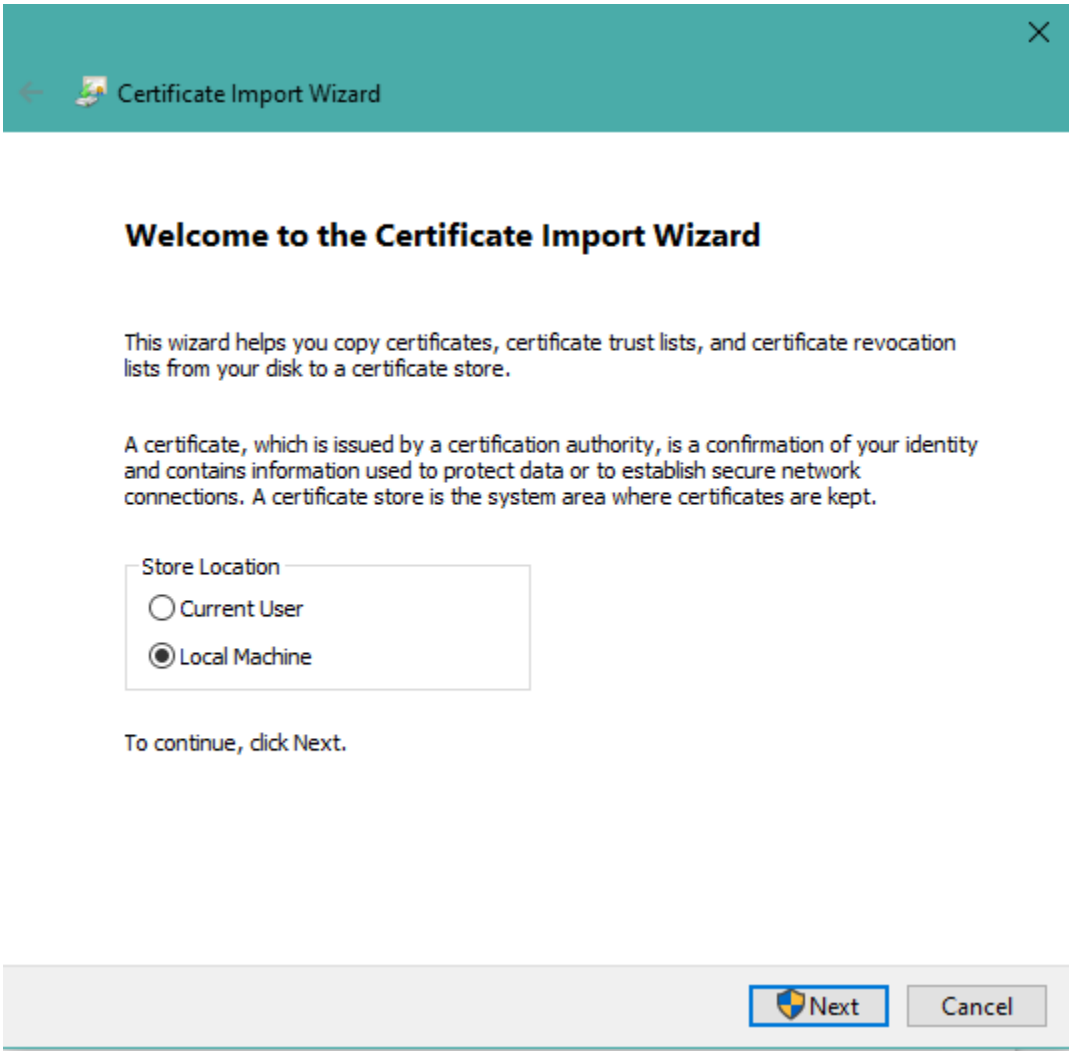
4.

In the certificate installer, select “Install Certificate



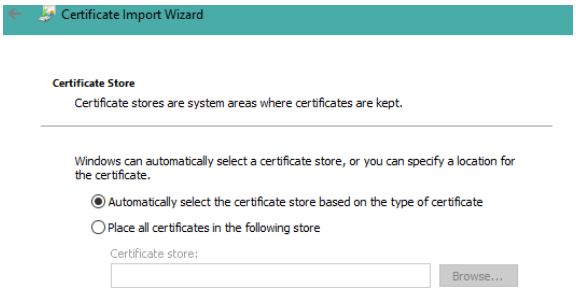
5.

Download the certificate to your local machine.



6.

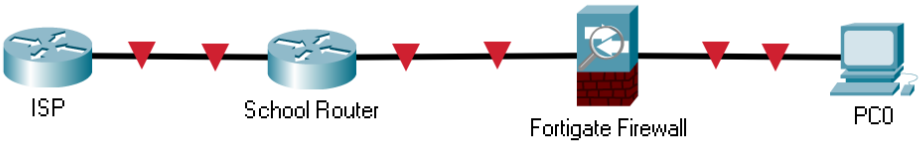
Automatically send the certificate to the correct folder. Then select finish.



Lab Summary

To apply our filters to the firewall, we first created our web filters in the security & objects > web filter, created a DNS filter in security & objects > DNS filter, and an application control filter in the security & objects > application control section. To apply these filters to the firewall, we created two firewall policies. One would allow traffic to be forwarded out onto the internet from our LAN and one would allow traffic to be forwarded into our LAN from the WAN. Then we applied all 3 filters on each security policy to begin blocking traffic. Next, we noticed that the certificate error was beginning to remove our block messages and give us untrusted certificate errors instead. Thus, we downloaded the certificate off the GUI by entering Security & objects > ssl

Topology



Problems

Didn't block traffic coming back in

> At first, we only created a traffic rule going outbound towards the internet, however this resulted in some websites not being blocked because we were not blocking information coming into our LAN, only blocking information going out.

Confusion on whether Web Filter is blocking HTTPS

> When accessing HTTPS websites, they return a certificate. However, what Fortigate does is replace the website's certificate with our own certificate. This means our comptuer doesn't trust our own certificate, thus we end up with a certificate error.

Conclusion

Fortigate offers a very versatile and simple filtering method for not just schools, but for security purposes as well. The simplicity of creating these policies combined with the

protection it offers can make it a very effective and simple solution to any network that requires these filters (like our school filtering situation). While Fortigate may not be the strongest in its firmware, it makes up for it in the networking capabilities it can offer to IT teams with simple management and effective deployment. Not to mention the ease in setting up the Fortigate SOHO network, being one of its main strong suits over other firewall brands such as Palo Alto. The ease in deployment in Fortigate and its effectiveness in networking definitely makes this firewall an excellent choice for majority of situations where granular control over networks are necessary.

Fortigate Web Filter

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Period 5

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